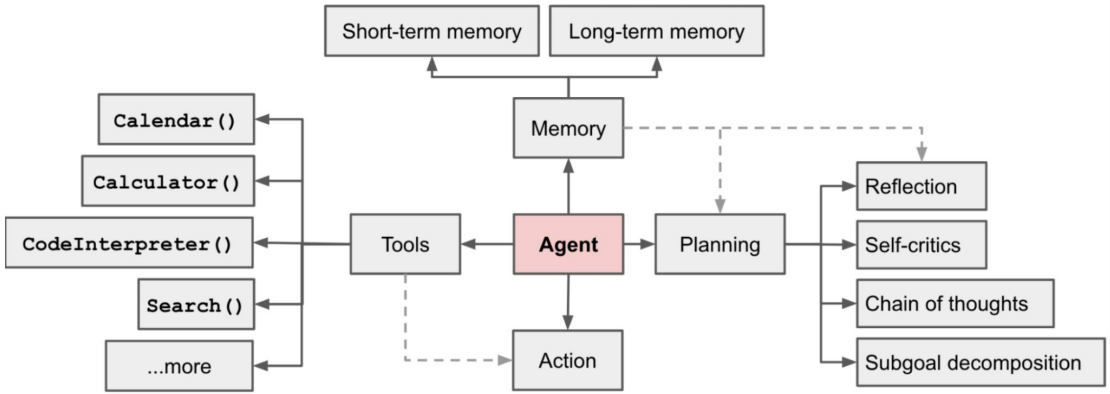
IV. Beyond LLM, Tools and (Multi)-Agents

A. **Beyond LLMs**
B. **Agents & Tools**
C. **Multi Agents Systems**



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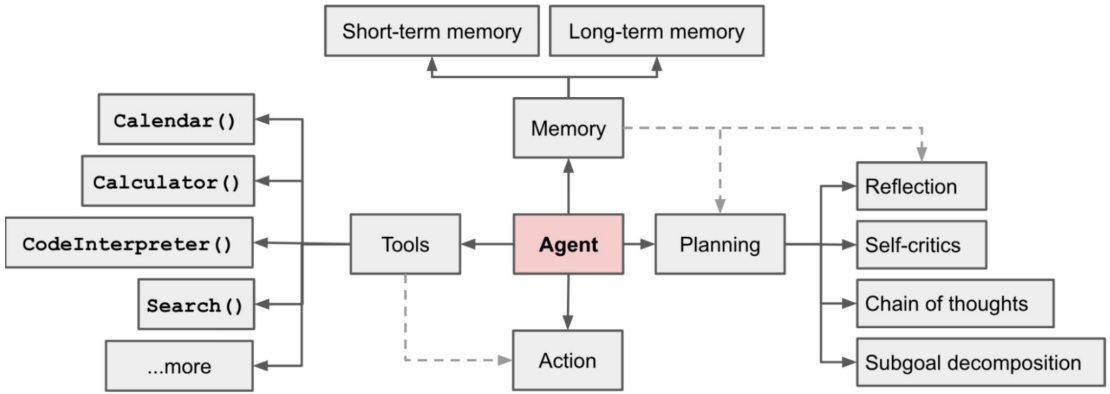
IV.A. Beyond LLMs

A. **Beyond LLMs**
B. **Agents & Tools**
C. **Multi Agents Systems**



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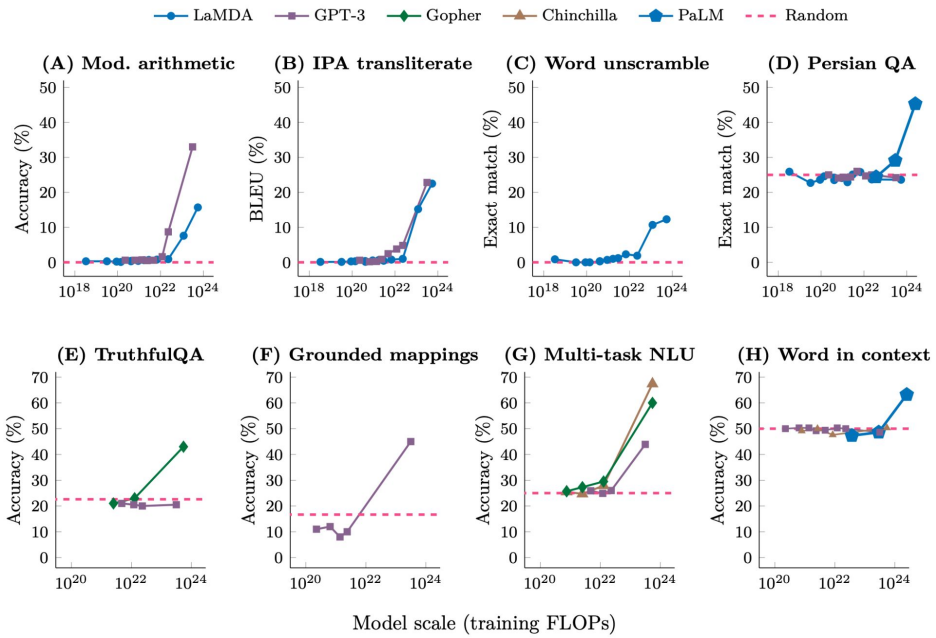
Back in the past

Emergent abilities definition:
An ability is emergent if it is present in larger but no smaller models

Performance is near random until a certain critic threshold
Know as “phase transition”



Wei & Al, 2022, Emergent Abilities of Large Language Models

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Table 1: List of emergent abilities of large language models and the scale (both training FLOPs and number of model parameters) at which the abilities emerge.

	Emergent scale			
	Train. FLOPs	Params.	Model	Reference
Few-shot prompting abilities				
• Addition/subtraction (3 digit)	2.3E+22	13B	GPT-3	Brown et al. (2020)
• Addition/subtraction (4-5 digit)	3.1E+23	175B		
• MMLU Benchmark (57 topic avg.)	3.1E+23	175B	GPT-3	Hendrycks et al. (2021a)
• Toxicity classification (CivilComments)	1.3E+22	7.1B	Gopher	Rae et al. (2021)
• Truthfulness (Truthful QA)	5.0E+23	280B		
• MMLU Benchmark (26 topics)	5.0E+23	280B		
• Grounded conceptual mappings	3.1E+23	175B	GPT-3	Patel & Pavlick (2022)
• MMLU Benchmark (30 topics)	5.0E+23	70B	Chinchilla	Hoffmann et al. (2022)
• Word in Context (WiC) benchmark	2.5E+24	540B	PaLM	Chowdhery et al. (2022)
• Many BIG-Bench tasks (see Appendix E)	Many	Many	Many	BIG-Bench (2022)
Augmented prompting abilities				
• Instruction following (finetuning)	1.3E+23	68B	FLAN	Wei et al. (2022a)
• Scratchpad: 8-digit addition (finetuning)	8.9E+19	40M	LaMDA	Nye et al. (2021)
• Using open-book knowledge for fact checking	1.3E+22	7.1B	Gopher	Rae et al. (2021)
• Chain-of-thought: Math word problems	1.3E+23	68B	LaMDA	Wei et al. (2022b)
• Chain-of-thought: StrategyQA	2.9E+23	62B	PaLM	Chowdhery et al. (2022)
• Differentiable search index	3.3E+22	11B	T5	Tay et al. (2022b)
• Self-consistency decoding	1.3E+23	68B	LaMDA	Wang et al. (2022b)
• Leveraging explanations in prompting	5.0E+23	280B	Gopher	Lampinen et al. (2022)
• Least-to-most prompting	3.1E+23	175B	GPT-3	Zhou et al. (2022)
• Zero-shot chain-of-thought reasoning	3.1E+23	175B	GPT-3	Kojima et al. (2022)
• Calibration via P(True)	2.6E+23	52B	Anthropic	Kadavath et al. (2022)
• Multilingual chain-of-thought reasoning	2.9E+23	62B	PaLM	Shi et al. (2022)

Wei & Al, 2022, Emergent Abilities of Large Language Models

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Increase model abilities

- Further scaling to train larger model with new emergent abilities
- New architecture, higher quality data, new training procedure could enable emergent abilities
- Further research to make the abilities available for small models

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Emergent Abilities limitations

- Societal risk with toxicity, bias and truthfulness
- Limitations with scaling LLM: Training data memorization (“overfitting”), toxicity and bias
- Other limitations we don’t have discovered yet

Sociological limitations

- Adoption of LLM, use of these models
- LLM used for a large range of tasks
  - Text
  - Image
  - Audio
  - Classification
  - …

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The future of Large Language Models:

- Generalist Agents
- Longer video understanding and generation
- Incredibly long sequence modeling (GPT authors a novel)
- Domain specific foundation models (Doctor GPT, Lawyer GPT)
- Real world impacts
  - Personalized education and tutoring systems
  - Advanced healthcare diagnostics
  - Real time multilingual communication

The New England Journal of Medicine is a registered trademark of [QA("Who is the publisher of The New England Journal of Medicine?") → Massachusetts Medical Society] the MMS.

Out of 1400 participants, 400 (or [Calculator(400 / 1400) → 0.29] 29%) passed the test.

The name derives from "la tortuga", the Spanish word for [MT("tortuga") → turtle] turtle.

The Brown Act is California's law [WikiSearch("Brown Act") → The Ralph M. Brown Act is an act of the California State Legislature that guarantees the public's right to attend and participate in meetings of local legislative bodies.] that requires legislative bodies, like city councils, to hold their meetings open to the public.

Shick & Al. 2023, Toolformer: Language Models Can Teach Themselves to Use Tools

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The future (What is missing):

Missing fonctionnalites to Artificial General Intelligence (AGI)

- Reducing computation complexity
- Enhance human controllability
- Adaptive Learning
- Infinite external memory
- Long horizon decision making
- Multi-sensory embodiment (physics and commonsense)

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Shick & Al. 2023, Toolformer: Language Models Can Teach Themselves to Use Tools

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Chain-of-Thought (CoT)

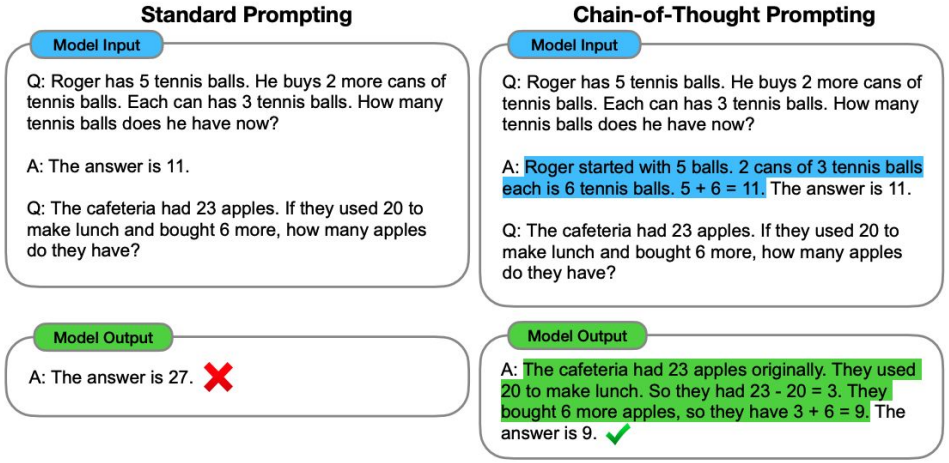
Definition: 'a series of intermediate reasoning steps significantly improves the ability of large language models to perform complex reasoning'

Example with one shot prompting:

"If Alice has 5 apples, and she buys 3 more, how many apples does she have in total?"

Without Chain-of-Thought (direct answer):
"8"

With Chain-of-Thought:
"Alice starts with 5 apples. She buys 3 more apples. So, 5 + 3 equals 8. Therefore, Alice has 8 apples."



Wei & Al, 2023, Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

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Chain-of-Thought (CoT)

CoT results in performance gains for larger LMs, but still have a non negligible fraction of errors

CoT works for 100B + params models
CoT fail for smaller models: “one step missing” or “semantic understanding” CoT errors are the most commons on smaller models

3 potentials reasons:

- Weaker arithmetic abilities
- Logical loopholes and do not arrive at final answer

New research area: Generalizing CoT for small models

Wei & Al, 2023, Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

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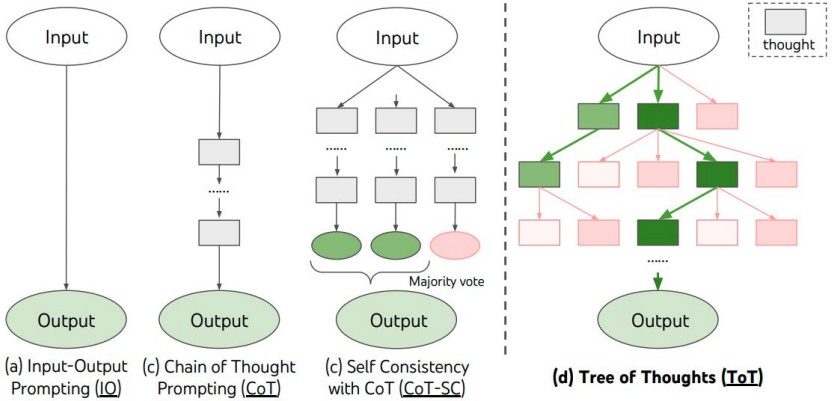
Tree of Thought (ToT)

Definition: ‘Consider multiple different reasoning paths and self-evaluation choices to decide the next course of action, as well as looking ahead or backtracking when necessary to make global choices’



Yao & Al. 2024. Tree of Thoughts: Deliberate Problem Solving with Large Language Models

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Program-Aided Language Model (PAL)

PAL add an intermediate reasoning step to transform a problem into Python code.

PAL distinguishes **reasoning** and **solving** in two separate steps

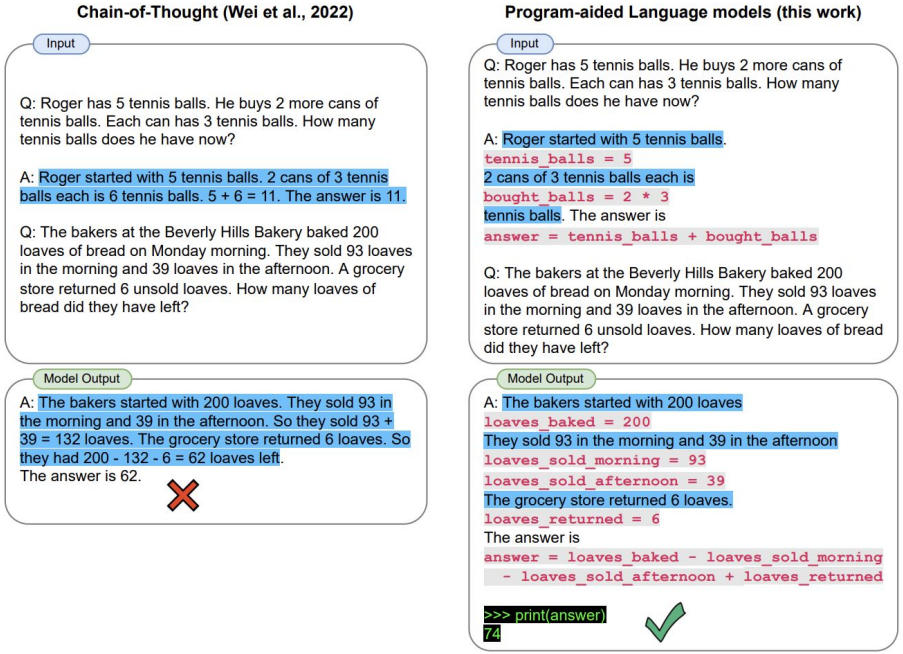

Figure 1: A diagram illustrating PAL: Given a mathematical reasoning question, Chain-of-thought (left) generates intermediate reasoning steps of free-form text. In contrast, Program-aided Language models (PAL, right) generate intermediate steps *and* Python code. This shifts the role of *running* the reasoning steps from the language model to the Python interpreter. The final answer is obtained by running the generated reasoning chain. Chain-of-thought reasoning is highlighted in blue; PAL steps are `highlighted in gray and pink`; the Python interpreter run is highlighted in black and green.

Gao & Al, 2022, PAL: Program-aided Language Models

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Program-Aided Language Model (PAL)

```
Q: Jane was born on the last day of Feburary in 2001. Today is her 16-year-old
↪  birthday. What is the date 24 hours later in MM/DD/YYYY?

A: The last day of February is the 28th, so Jane was born on 02/28/2001. Today is her
↪  16-year old birthday, so today is 02/28/2017. So 24 hours later is 02/29/2017. So
↪  the answer is 02/29/2017.
```

```
# Q: Jane was born on the last day of Feburary in 2001. Today is her 16-year-old
    birthday. What is the date 24 hours later in MM/DD/YYYY?
# If Jane was born on the last day of Feburary in 2001 and today is her 16-year-old
    birthday, then today is 16 years later.
today = datetime(2001, 2, 28) + relativedelta(years=16)
# 24 hours later,
later = today + relativedelta(hours=24)
# The answer formatted with %m/%d/%Y is
later.strftime('%m/%d/%Y')
```

Figure 25: Example model generation on Date Understanding.

Gao & Al, 2022, PAL: Program-aided Language Models

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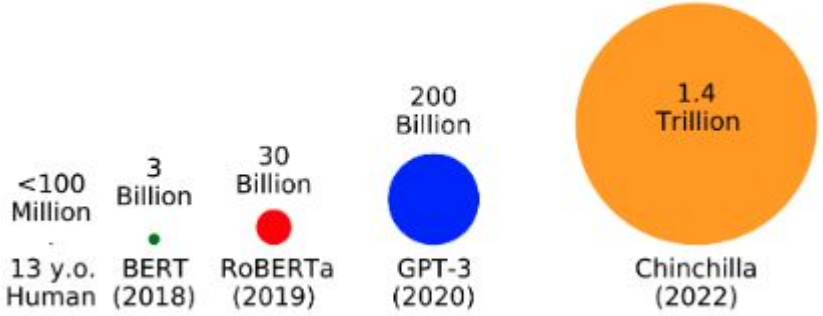
Baby LM: Children vs LLMs

- A LM sees $10^{4\text{-}5}$ more words than a 4 years old child
- Children learn in a smarter, more explicit compositional/hierarchical manners learning abstraction/generalization and reasoning more easily
- Baby LM Challenge: train a powerful model with just a limited amount of data
- LM on device (edge computing, i.e smartphones



Baby LM Challenge

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Baby LM: Children vs LLMs

Developmentally plausible pre training dataset inspired by the input to children

- Under 100M words: Children are exposed to 2M-7M words per year (Gilkerson et al., 2017). Choosing the beginning of adolescence (age 13) as a cutoff, children are exposed to at maximum 91M words, which they round up to 100M.
- Mostly transcribed speech: Most of the input to children is spoken; thus, the dataset focuses on transcribed speech.
- Mixed domain, consisting of the following sources: CHILDES (child-directed speech), OpenSubtitles (speech), BNC (speech), TED talks (speech), children's books (simple written language), Wikipedia, and Simple Wikipedia.

Baby LM Challenge

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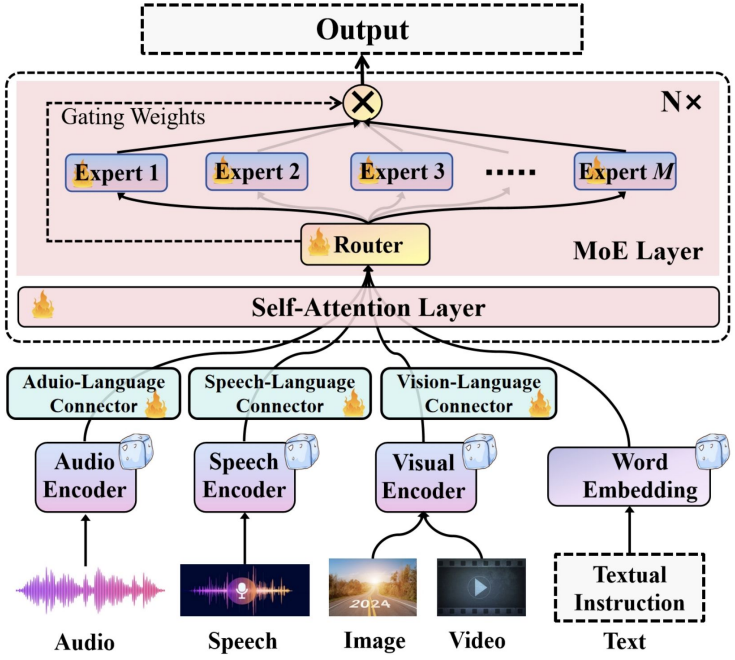
Mixture of Expert (MoE) model

- Combination of multiple small Neural networks knowns as expert which are trained and capable of handling particular data and performing specialised task
- Gatin network predicts which response is best suited to address the request



Li & Al, 2024.5 Uni-MoE: Scaling Unified Multimodal LLMs with Mixture of Experts

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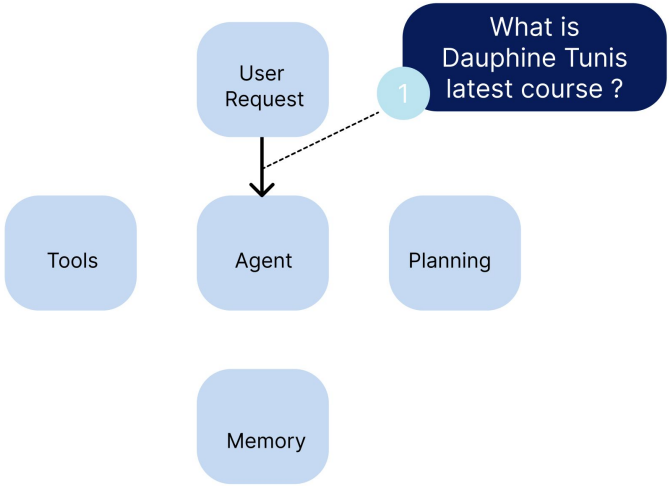
IV.B. Agents & Tools

A. **Beyond LLMs**
B. **Agents & Tools**
C. **Multi Agents Systems**



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Example

https://www.youtube.com/watch?v=Ubr7FWmeIyM&t=75s

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Why Agents ?

1. **Why ?** A single LLM call is not enough to do complex tasks
2. **How ?** Using model chaining, reflection, chain of thought …
3. **Tools ?** Memory, context length, personalization, actions, internet access…

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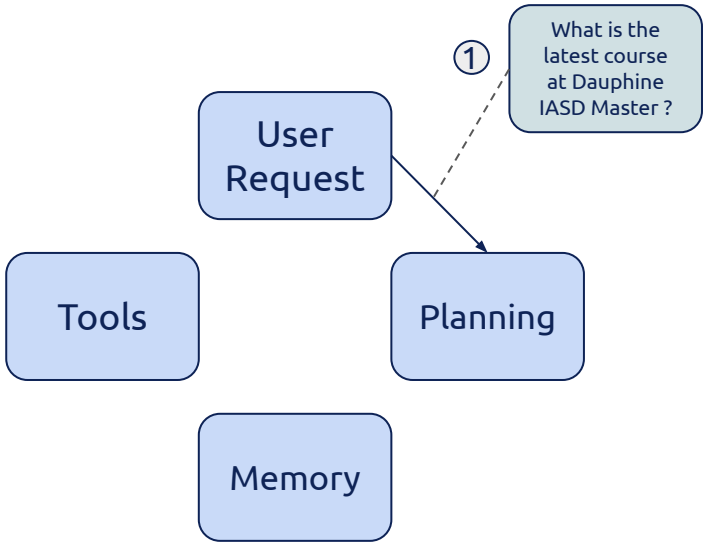
Agents example



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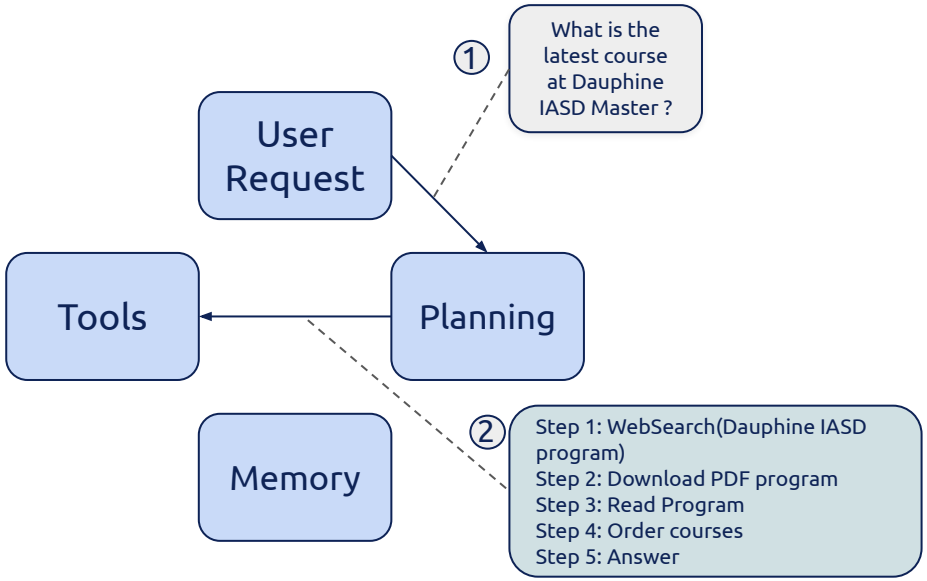
Agents example



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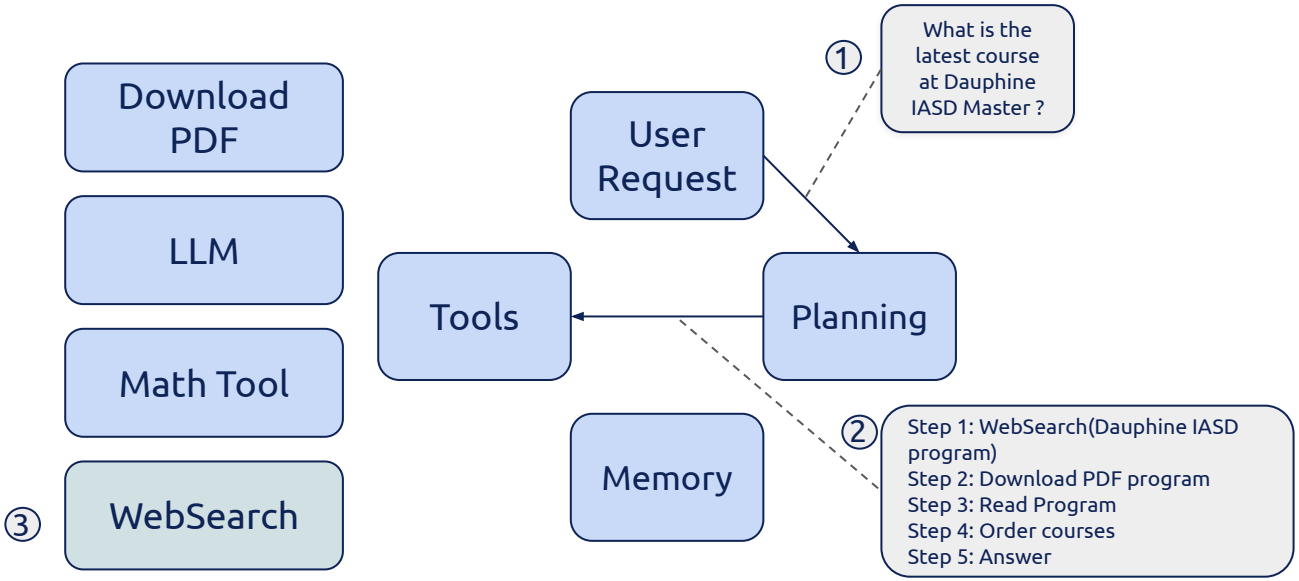
Agents example



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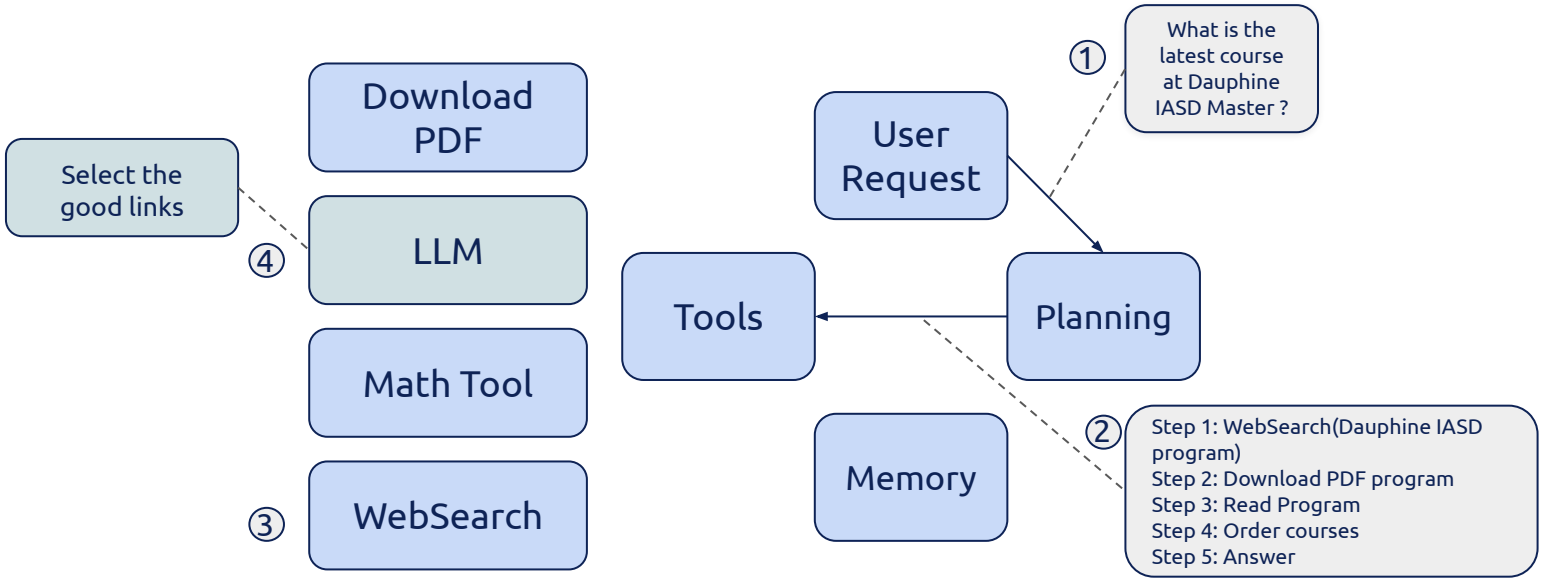
Agents example



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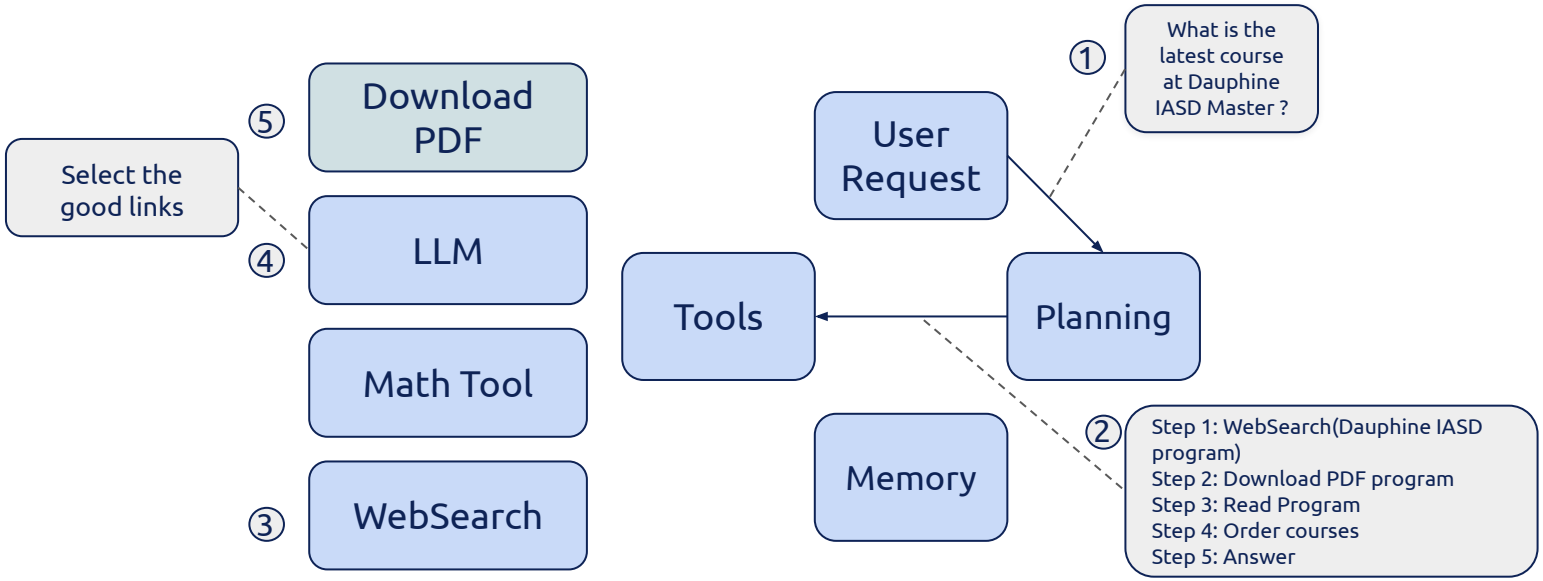
Agents example



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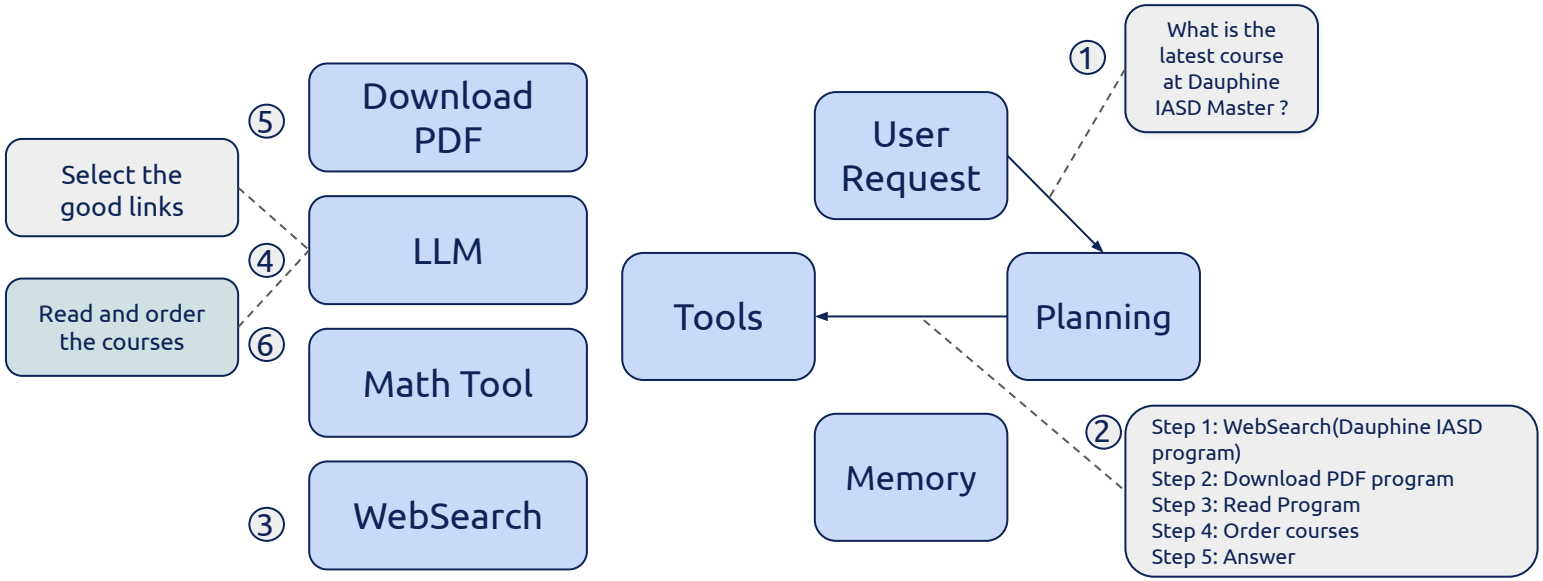
Agents example



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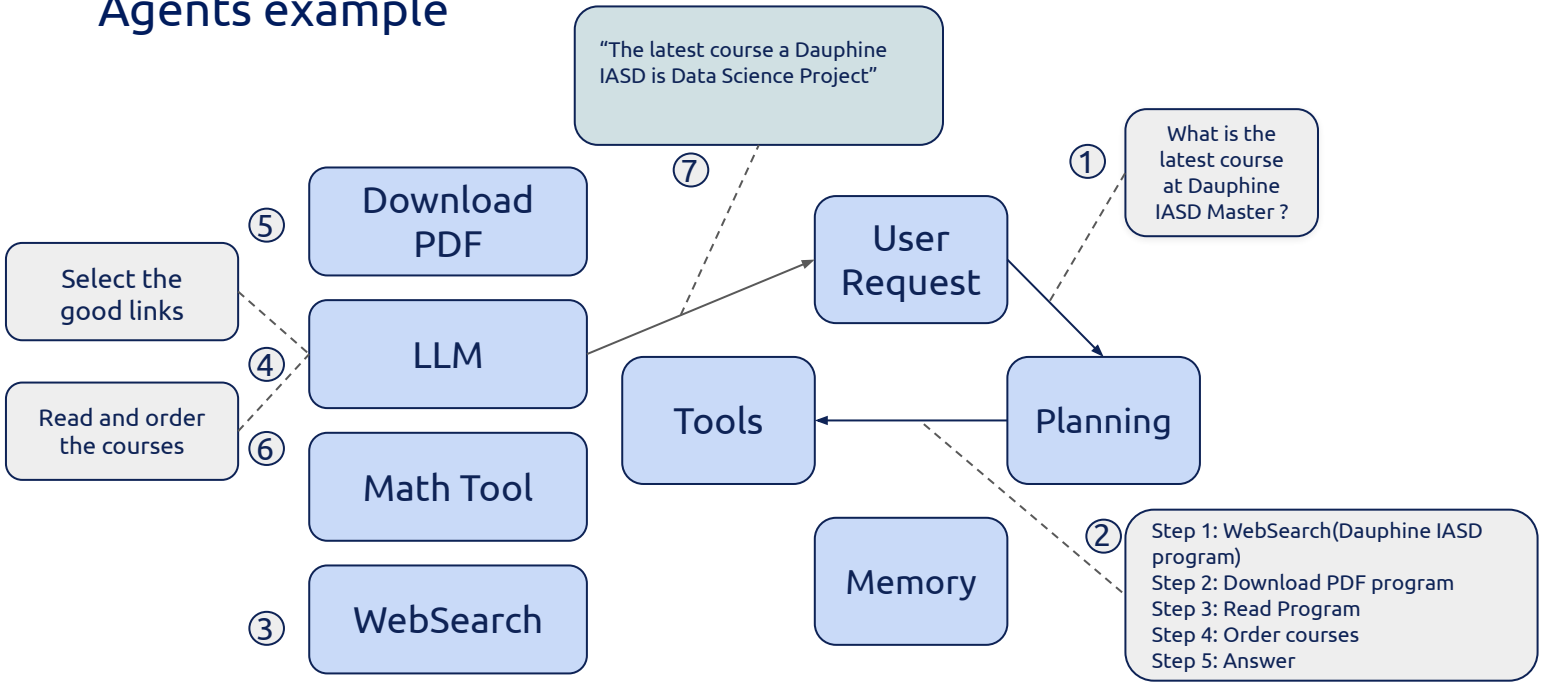
Agents example



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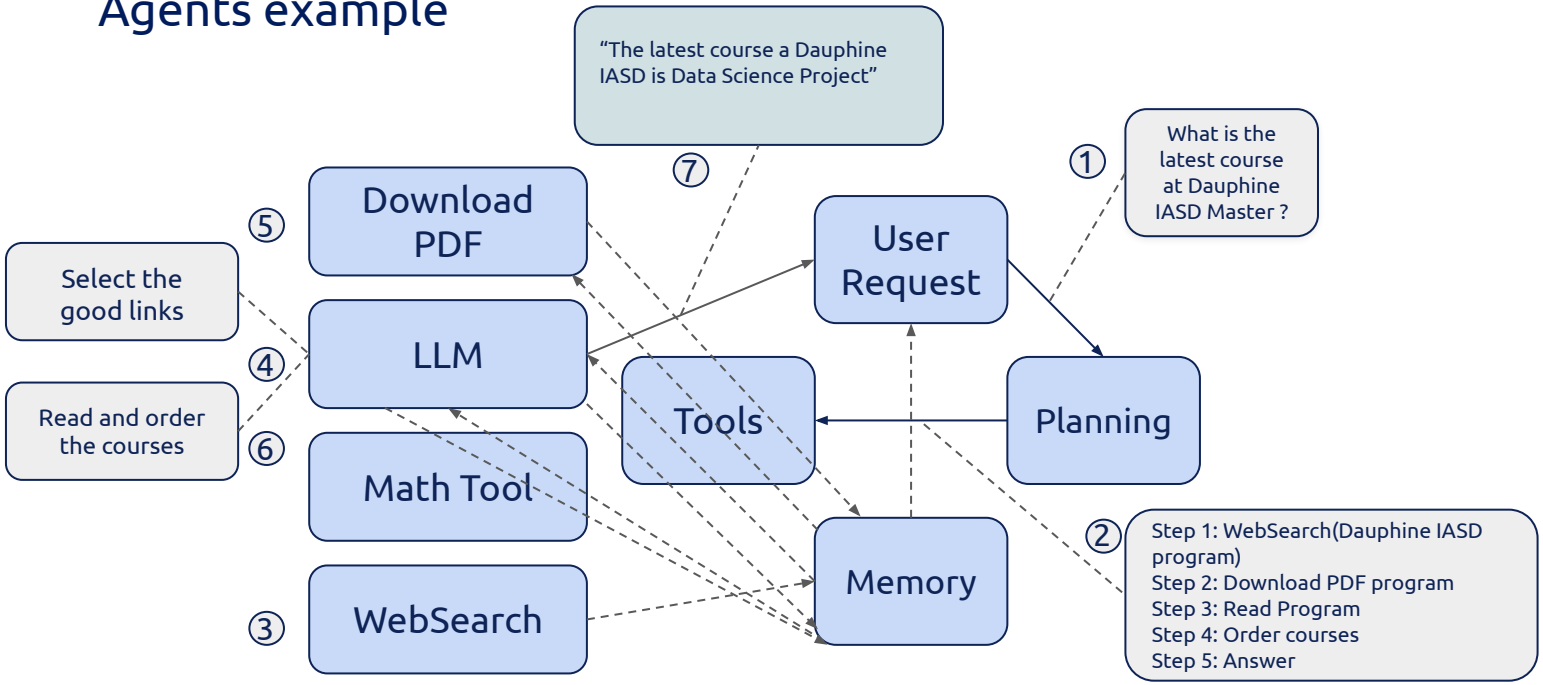
Agents example



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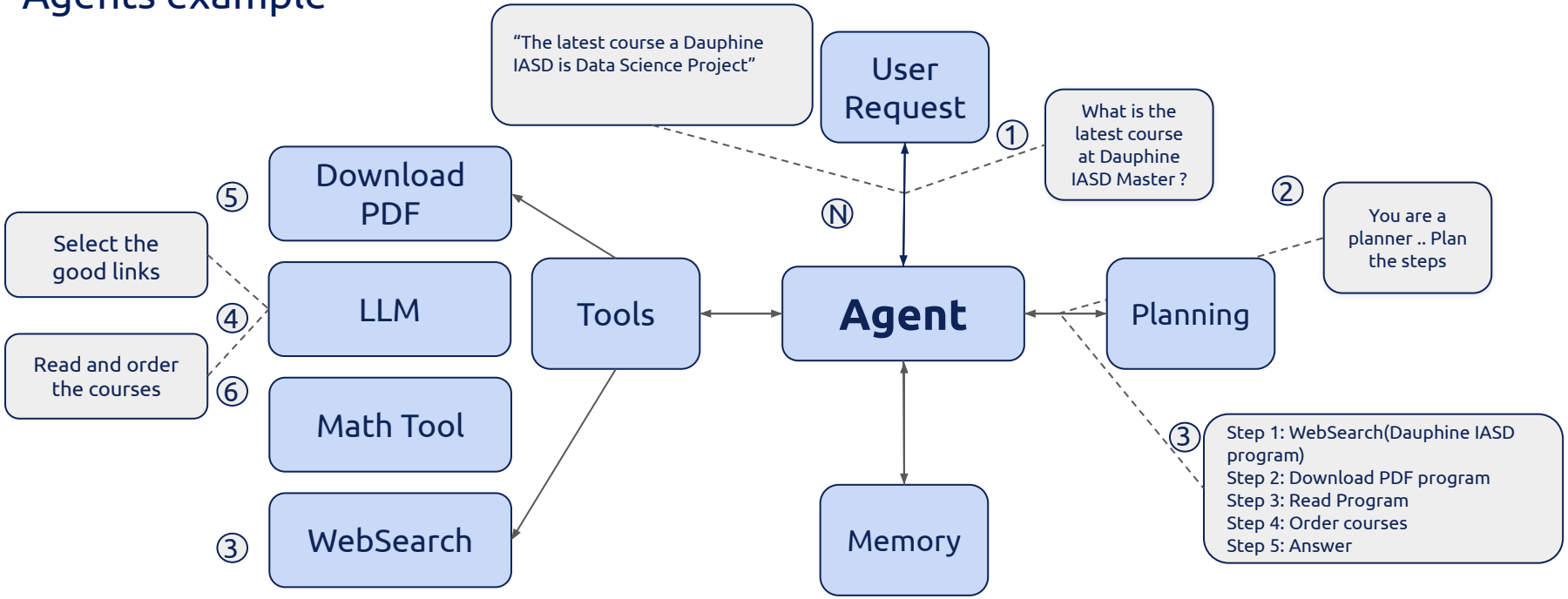
Agents example



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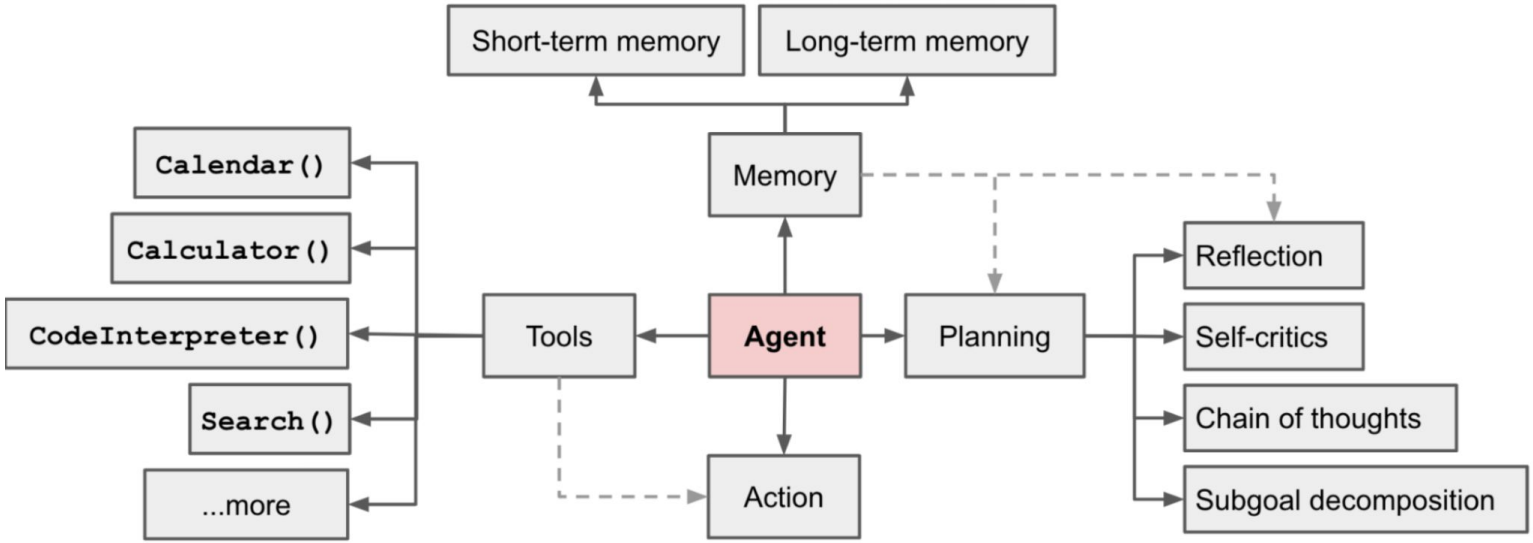
Agents



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Human like AI Agents

Building agents with LLM (large language model) as its core controller is a end of 2023 concept.

Several proof-of-concepts demos, such as:

- AutoGPT, 167K ⭐
- GPT-Engineer, 52K ⭐
- ChatDev, 25K ⭐
- BabyAGI, 20K ⭐

are Agent example.

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Chat Dev

ChatDev, 25K ⭐

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Human like Agents

1. **Capable of using human-designed interfaces**: Able to work with existing interfaces meant for humans, transcending typical programmatic limitations.
2. **Acts as a digital extension**: Functions as an extension of the user, taking actions on their behalf.
3. **Fewer restrictions**: Can manage logins, payments, and interact with services without being limited by API constraints.
4. **Simple actions required**: Operates using basic action primitives, such as clicking and typing.
5. **Learns and improves**: Continuously learns from user interactions and enhances its performance over time.

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Artificial general intelligence (AGI)

Definition: type of artificial intelligence (AI) that matches or surpasses human cognitive capabilities across a wide range of cognitive tasks.

To create an AGI, AI systems have to perform better than human.

Q. How to make performance comparison ?

Leaderboard

Submission made by our team are labelled "GAIA authors". While we report average scores over different runs when possible in our paper, we only report the best run in the leaderboard.

See below for submissions.

Citation

Results: Test | Results: Validation

Agent name	Average score (%)	Level 1 score (%)	Level 2 score (%)	Level 3 score (%)	organisation	Model family	Submission date
Multi Agent	38.87	53.76	37.74	14.29			2024-10-10
das_agent v0.4	38.21	51.61	36.48	18.37	NA	GPT-4o	2024-10-04
Trase Agent v0.1	35.55	50.54	33.33	14.29	Trase Systems	Fine-tuned GPT-4o	2024-09-04
Sibyl System v0.2	34.55	47.31	32.7	16.33	Baichuan Inc.	GPT-4o	2023-11-03

Hugging Face, General AI Assistant Benchmark

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General AI Assistant benchmark (GAIA)

We **evaluate** the systems by their capacity of solving **tasks that requires reasoning**
We **don't evaluate** if they reason **like** humans

GAIA: A Benchmark for General AI Assistants

Grégoire Mialon[1], **Clémentine Fourrier**[2], **Craig Swift**[3], **Thomas Wolf**[2], **Yann LeCun**[1], **Thomas Scialom**[4]

[1]FAIR, Meta, [2]HuggingFace, [3]AutoGPT, [4]GenAI, Meta

We introduce GAIA, a benchmark for General AI Assistants that, if solved, would represent a milestone in AI research. GAIA proposes real-world questions that require a set of fundamental abilities such as reasoning, multi-modality handling, web browsing, and generally tool-use proficiency. GAIA questions are conceptually simple for humans yet challenging for most advanced AIs: we show that human respondents obtain 92% vs. 15% for GPT-4 equipped with plugins. This notable performance disparity contrasts with the recent trend of LLMs outperforming humans on tasks requiring professional skills in *e.g.* law or chemistry. GAIA's philosophy departs from the current trend in AI benchmarks suggesting to target tasks that are ever more difficult for humans. We posit that the advent of Artificial General Intelligence (AGI) hinges on a system's capability to exhibit similar robustness as the average human does on such questions. Using GAIA's methodology, we devise 466 questions and their answer. We release our questions while retaining answers to 300 of them to power a leader-board hereby accessible.

Date: November 23, 2023
Correspondence: {gmialon,tscialom}@meta.com, clementine@huggingface.co
Code: https://huggingface.co/gaia-benchmark

Level 1

Question: What was the actual enrollment count of the clinical trial on H. pylori in acne vulgaris patients from Jan-May 2018 as listed on the NIH website?
Ground truth: 90

Level 2

Question: If this whole pint is made up of ice cream, how many percent above or below the US federal standards for butterfat content is it when using the standards as reported by Wikipedia in 2020? Answer as + or - a number rounded to one decimal place.
Ground truth: +4.6

Level 3

Question: In NASA's Astronomy Picture of the Day on 2006 January 21, two astronauts are visible, with one appearing much smaller than the other. As of August 2023, out of the astronauts in the NASA Astronaut Group that the smaller astronaut was a member of, which one spent the least time in space, and how many minutes did he spend in space, rounded to the nearest minute? Exclude any astronauts who did not spend any time in space. Give the last name of the astronaut, separated from the number of minutes by a semicolon. Use commas as thousands separators in the number of minutes.
Ground truth: White; 5876

LeCun, Wolf & Al, 2023, GAIA: A Benchmark for General AI Assistants

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Will AI Systems become AGI ?

2023 Nov	2023 March	2024 Oct	202X ?
GAIA paper	**Auto Gen**	**“Multi Agent”**	**AGI ?**
Human: 92%	Human: 92%	Human: 92%	Human: 92%
AI System: 13%	AI System: 34%	AI System: 38.87%	AI System: ?

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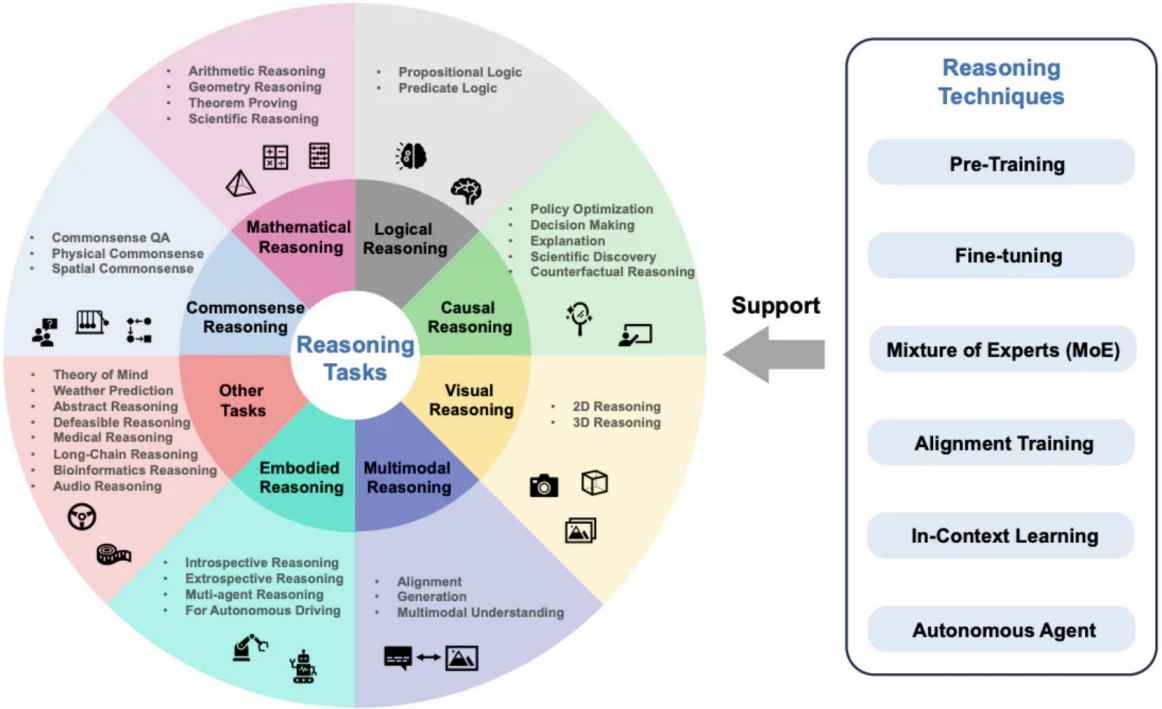
Reasoning tasks challenges



Sun & Al, 2023, Reasoning with Language Model Prompting: A Survey

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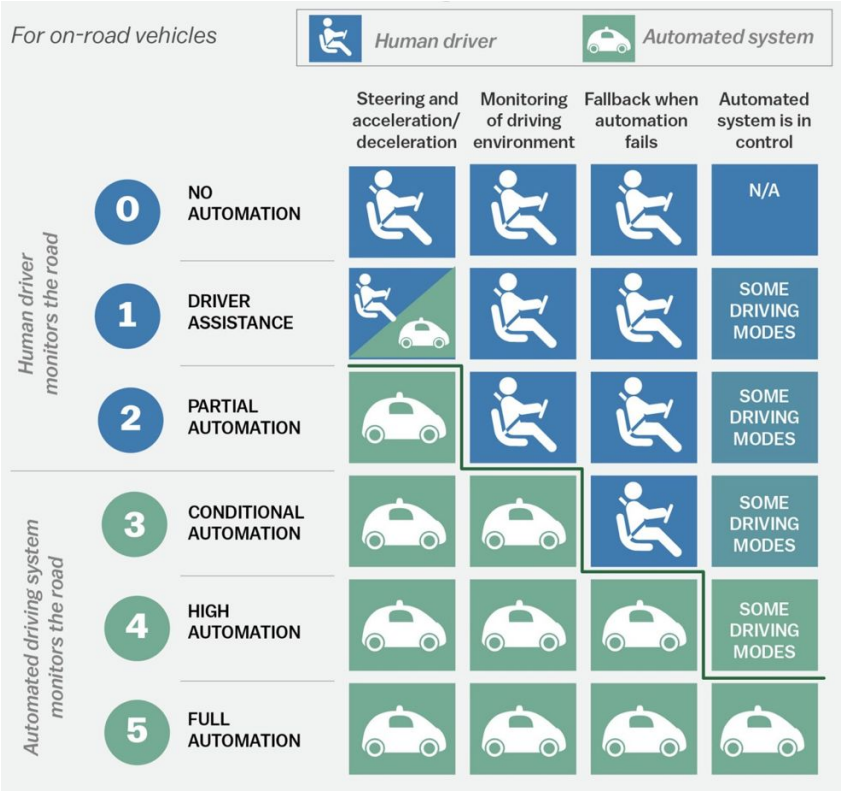
Automation Levels



The Rise of Autonomous Vehicles

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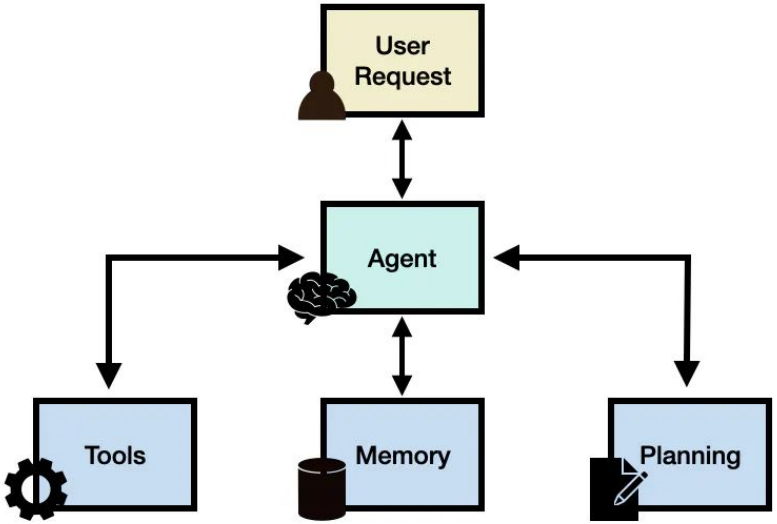
Agent components

Generally speaking, an LLM agent framework can consist of the following core components:

User Request: a user question or request
Agent/Brain: the agent core acting as coordinator
Planning: assists the agent in planning future actions
Memory: manages the agent's past behaviors



Wang & Al. 2024. A Survey on Large Language Model based Autonomous Agents

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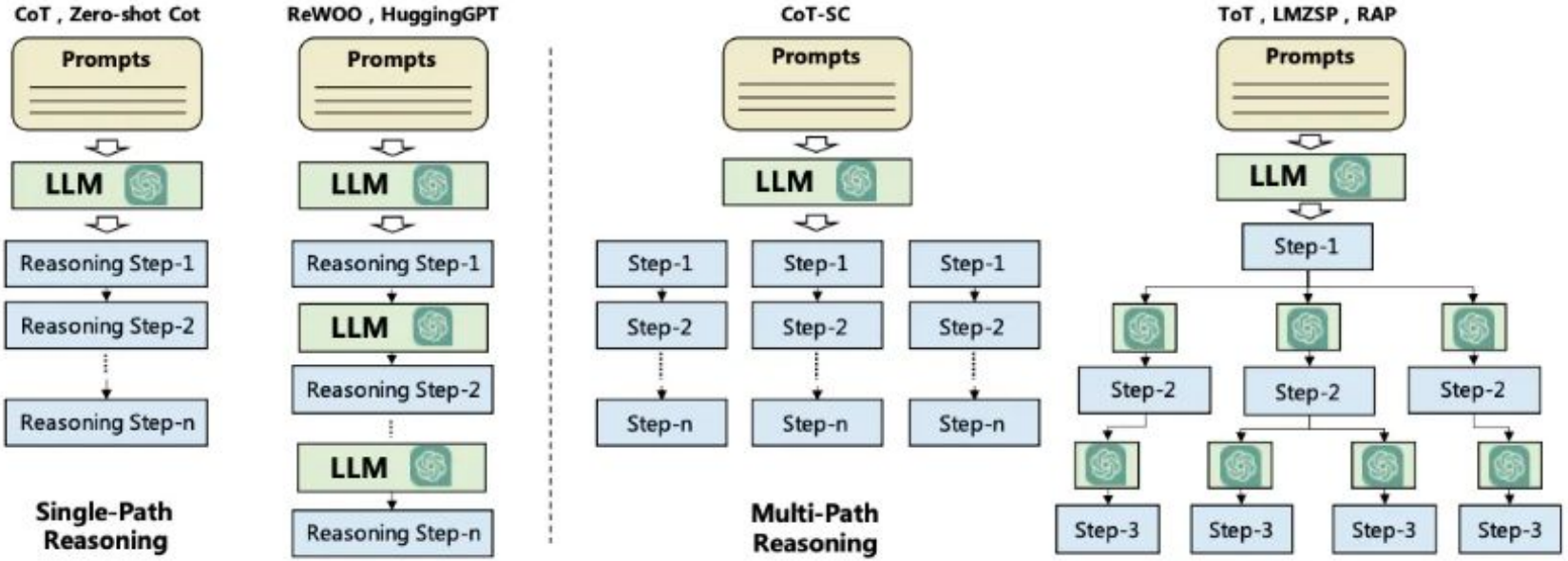
Planning strategies



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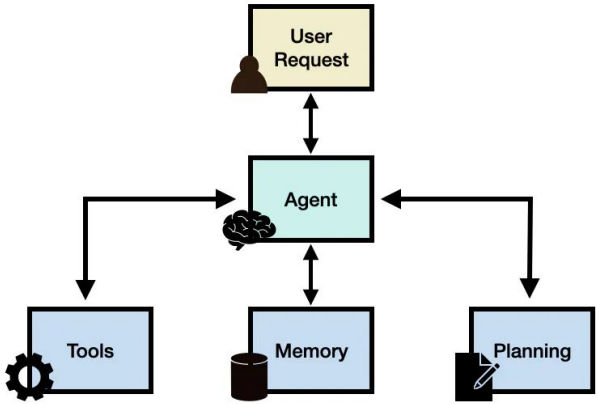
Planning feedback

Planning module have no feedback
The plan is provided at the beginning of the execution and will fail if any of the step fails

Q. How to address this challenge ?



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Synergizing Reasoning and Acting in Language Models (ReAct)

(1) Hotspot QA

Question: Aside from the Apple Remote, what other device can control the program Apple Remote was originally designed to interact with?

(1a) Standard

Answer: iPod ✗

(1b) CoT (Reason Only)

Thought: Let's think step by step. Apple Remote was originally designed to interact with Apple TV. Apple TV can be controlled by iPhone, iPad, and iPod Touch. So the answer is iPhone, iPad, and iPod Touch.

Answer: iPhone, iPad, iPod Touch ✗

(1c) Act-Only

Act 1: Search[Apple Remote]
Obs 1: The Apple Remote is a remote control …

Act 2: Search[Front Row]
Obs 2: Could not find [Front Row]. Similar: …

Act 3: Search[Front Row (software)]
Obs 3: Front Row is a discontinued media center software …

Act 4: Finish[yes] ✗

(1d) ReAct (Reason + Act)

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.
Act 1: Search[Apple Remote]
Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple … originally designed to control the Front Row media center program …

Thought 2: Apple Remote was originally designed to control the Front Row media center program. I need to search Front Row next and find what other device can control it.
Act 2: Search[Front Row]
Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports',' Front Row (software)', …]

Thought 3: Front Row is not found. I need to search Front Row (software) .
Act 3: Search[Front Row (software)]
Obs 3: Front Row is a discontinued media center software …

Thought 4: Front Row (software) is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.
Act 4: Finish[keyboard function keys] ✓

Prompt Method[a]	HotpotQA (EM)	Fever (Acc)
Standard	28.7	57.1
CoT (Wei et al., 2022)	29.4	56.3
CoT-SC (Wang et al., 2022a)	33.4	60.4
Act	25.7	58.9
ReAct	27.4	60.9
CoT-SC → ReAct	34.2	**64.6**
ReAct → CoT-SC	**35.1**	62.0
Supervised SoTA[b]	67.5	89.5

Yao & Al, 2023. Synergizing Reasoning and Acting in Language Models (ReAct)

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Memory

Definition: The **memory** module is essential for storing an agent's internal logs, including previous thoughts, actions, and observations from interactions with the environment and users. In the literature on LLM agents, two primary types of memory have been identified:

- **Short-term memory**: This contains context about the agent's current situation, often implemented through in-context learning. It is limited and short-lived due to the constraints of the context window.
- **Long-term memory**: This stores the agent's past behaviors and thoughts that must be retained and accessed over longer periods. It typically relies on an external vector store, enabling fast and scalable retrieval of relevant information when needed.

A **hybrid memory** system combines both short-term and long-term memory to enhance the agent's capacity for long-range reasoning and accumulating experience.

Both the planning and memory modules allow the agent to operate in a dynamic environment and enable it to effectively recall past behaviors and plan future actions.

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Tools

Definition: Tools refer to a set of utilities that enable the LLM agent to interact with external environments, such as the Wikipedia Search API, Code Interpreter, or Math Engine.

These tools can also include:

- databases
- knowledge bases
- external models.

When the agent uses these external tools, it carries out tasks through workflows designed to help the agent gather observations or relevant information needed to complete subtasks and fulfill user requests.

Example:
In a health-related query, a code interpreter serves as a tool that executes code and generates the chart information requested by the user.

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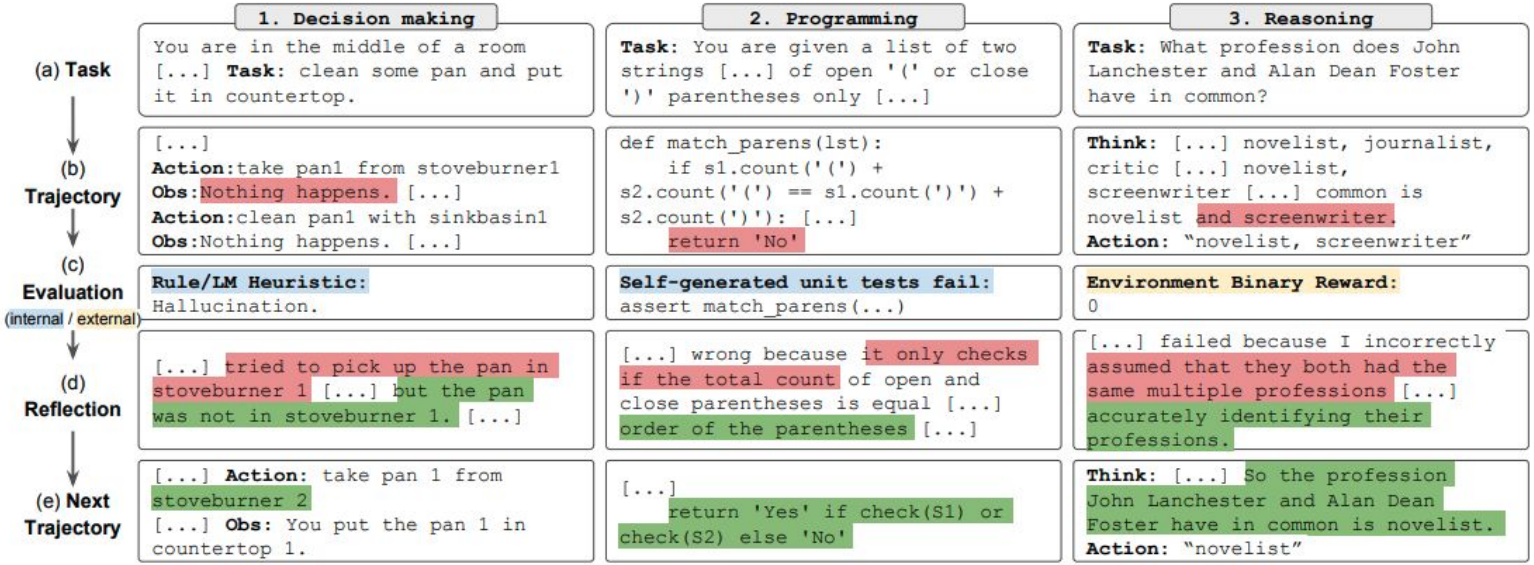
Reflexion: Language Agents with Verbal Reinforcement Learning



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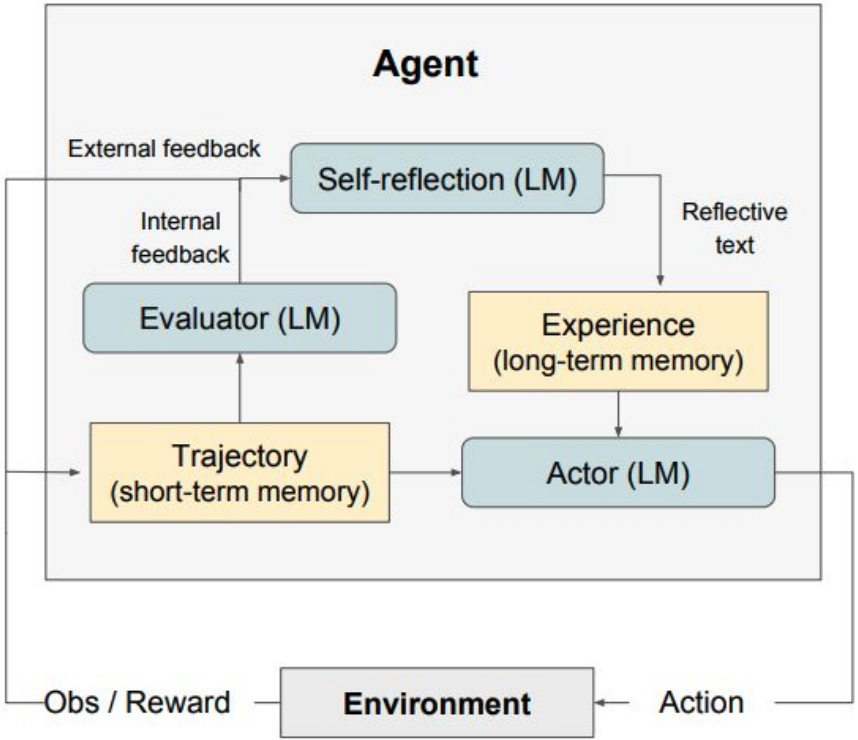
Reflexion: Language Agents with Verbal Reinforcement Learning



Shinn & Al, 2023, Reflexion: Language Agents with Verbal Reinforcement Learning

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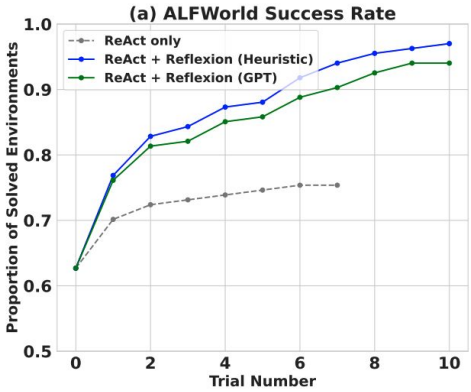
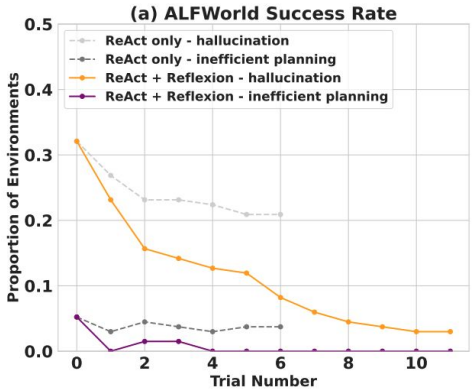
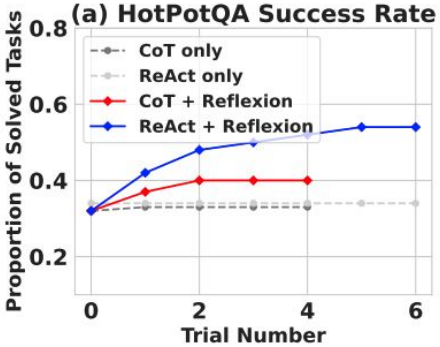
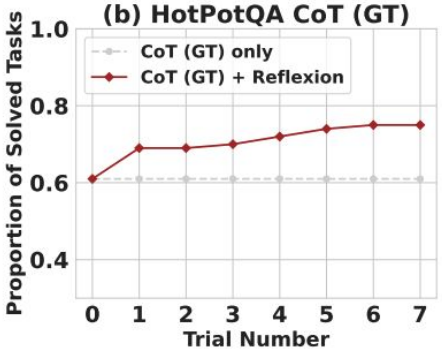
Reflexion: Language Agents with Verbal Reinforcement Learning

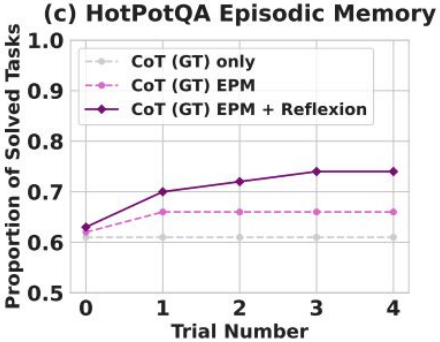
Performance











Shinn & Al, 2023, Reflexion: Language Agents with Verbal Reinforcement Learning

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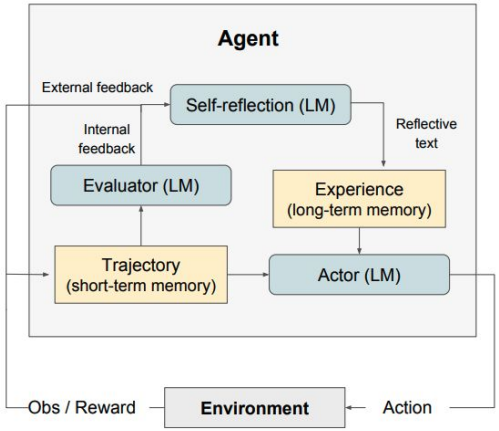
Reflexion: Language Agents with Verbal Reinforcement Learning

Definition: Reflexion is a framework designed to enhance language-based agents by using linguistic feedback.
Reflexion introduces a novel approach to verbal reinforcement, where the policy is defined by the agent's memory encoding combined with selected LLM parameters.

- At its core, Reflexion transforms feedback whether in the form of free language or numeric scores from the environment into linguistic feedback, also known as self-reflection.
- This feedback is then incorporated as contextual information for the LLM agent in subsequent interactions.
- This process enables the agent to quickly learn from past errors, leading to improved performance across various complex tasks.



Shinn & Al. 2023. Reflexion: Language Agents with Verbal Reinforcement Learning

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Challenges

- Collecting user data & preferences:
  - Actively asking for preferences
  - Passive learning from interactions
- Learning from user preferences: Supervised fine-tuning vs human-feedback
- On-fly adaptation
- Privacy

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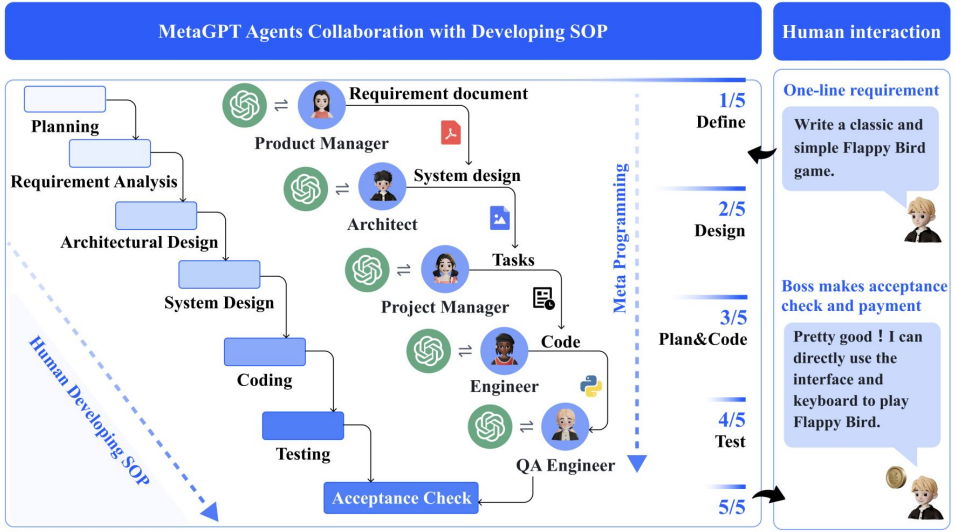
IV.C. Multi Agents Systems

A. Beyond LLMs
B. Agents & Tools
C. Multi Agents Systems



Hong & Al, 2023.11, Meta GPT

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Multi Agent Systems

Definition: LLM-based multi-agent systems refer to frameworks where multiple autonomous agents, each potentially powered by a Large Language Model, interact within a shared environment. These agents can cooperate, compete, or operate independently, aiming to achieve individual or shared goals.

- **Autonomy**: Each agent operates independently and can make decisions based on its objectives, knowledge, and the perceived state of the environment.
- **Scalability**: LLMs enable these agents to handle complex tasks and large amounts of data, making the system scalable to various applications and scenarios.
- **Interoperability**: Agents can communicate and negotiate with each other, using natural language or other forms of structured data facilitated by their underlying LLM capabilities.
- **Adaptability**: These systems are capable of adapting to new information or changes in the environment, learning from interactions and refining their strategies over time.

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Why Multi Agent Systems

Parallelization is possible for Multi Agents systems

Task specialization: A agent can be the intermediate between the user and services (spreadsheet, slides, internet expert, coder)

Challenges:
- Agent to Agent Communication

How to exchange informations (natural languages) ?
Like human, there are misunderstanding

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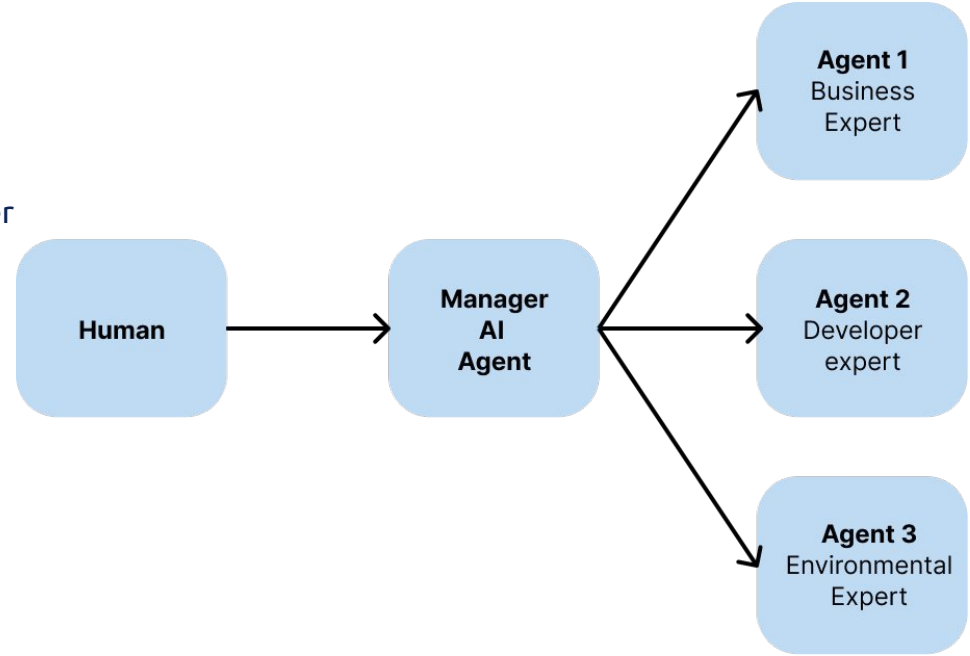
Agent to agent communication

- Agents can communicate between each other
- Hierarchy notion like manager
- Worker can run on different GPU
- Create failover mechanism



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Limitations

- **Reliability**: Ensuring that the autonomous agent consistently performs its tasks without failure or errors.
- **Looping and Plan Divergence**: Addressing scenarios where the agent's decision-making process leads to infinite loops or deviates from the original plan.
- **Testing and Benchmarking:** Developing robust methods to evaluate and benchmark the performance of autonomous agents across different tasks and environments.
- **Real-World Deployment and Observability:**
  - Trust: How do we ensure trust in fully autonomous AI systems, especially in high-stakes environments?
  - Human Overrides: How can we integrate mechanisms for human intervention or overrides in the event of unforeseen circumstances or errors ?

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Limitations

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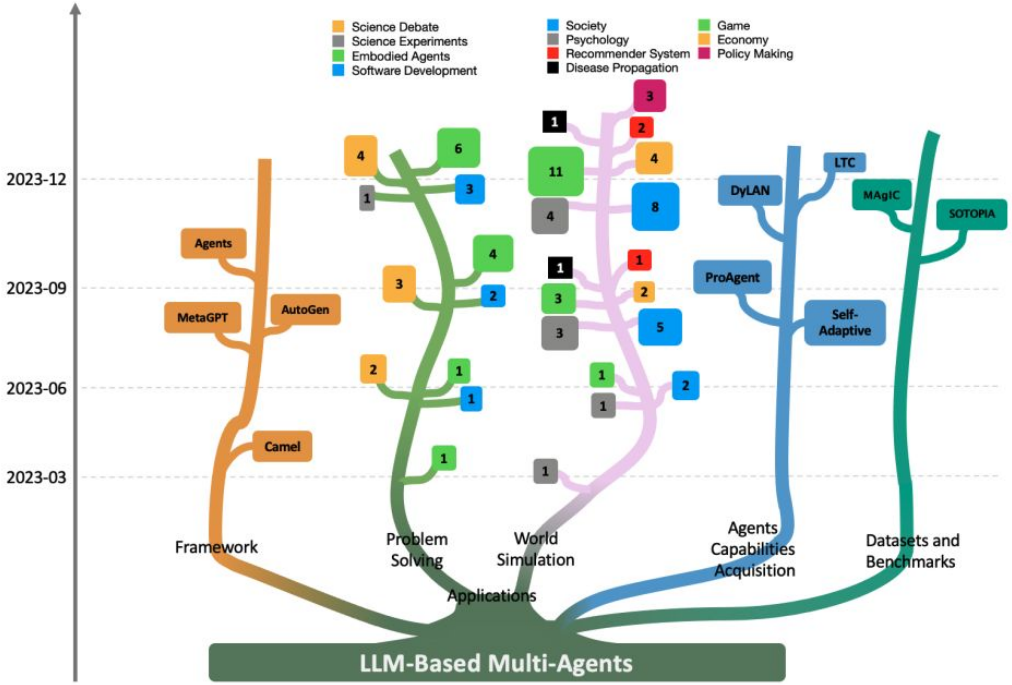
Survey of progresses and challenges



Guo & al. 2024, Large Language Model based Multi-Agents: A Survey of Progress and Challenges

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Multi Agent Systems

Y Combinator, 2024, October, Why Vertical LLM Agents Are The New $1 Billion SaaS Opportunities

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